



Hydroclimatic variability and trend analysis of the Arno River basin, Tuscany, Italy

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Abstract

Climate change is altering hydroclimatic regimes across the Mediterranean, yet recent basin-scale observational syntheses remain limited for key river basins. This study evaluates hydroclimatic variability and trends in the Arno River basin, Tuscany, Italy, using daily temperature, precipitation, and streamflow records for 1994–2023. Trends were assessed using autocorrelation-corrected modified Mann–Kendall tests, Sen’s slope estimates with 95% confidence intervals, Benjamini–Hochberg multiple-testing correction, and change-point diagnostics. Statistically significant warming was detected in summer ($+0.066\text{ }^{\circ}\text{C yr}^{-1}$) and autumn ($+0.073\text{ }^{\circ}\text{C yr}^{-1}$), both trends remaining significant after multiple-testing correction. Annual mean temperature increased by $+0.052\text{ }^{\circ}\text{C yr}^{-1}$, equivalent to $+1.56\text{ }^{\circ}\text{C}$ over 30 years, but did not remain significant after multiple-testing correction. A significant temperature change-point occurred around 2010, while no change-points were detected for precipitation or streamflow. No significant monotonic trends were detected for precipitation or streamflow at annual or seasonal scales, although the 30-year record and high interannual variability (CV up to $\sim 36\%$ for precipitation and $\sim 66\%$ for streamflow) limit exclusion of small trends. Precipitation displayed a clear orographic gradient, with higher values along the Apennine ridge and lower values in lowlands. Precipitation and streamflow were positively correlated across all seasons ($\rho=0.51\text{--}0.93$), indicating strong precipitation-runoff coupling. Summer temperature was negatively correlated with streamflow ($\rho = -0.53$, $p=0.003$; $\text{pBH}=0.009$), consistent with warm-season hydroclimatic sensitivity, although this observational association does not establish causation. These findings provide an updated observational baseline for hydroclimatic monitoring and climate-adaptation planning in the Arno River basin.

Highlights

- Summer and autumn temperature increased by $+0.066$ and $+0.073\text{ }^{\circ}\text{C yr}^{-1}$, respectively.
- Annual temperature increased by $+0.052\text{ }^{\circ}\text{C yr}^{-1}$, with a significant change point around 2010.
- Precipitation and streamflow showed no significant monotonic trends over 1994–2023.
- Precipitation–streamflow correlations $\rho=0.51\text{--}0.93$ indicate strong runoff coupling.
- Summer temperature–streamflow correlation $\rho = -0.53$ indicates warm-season sensitivity.

Keywords Climate change · Trend analysis · Modified Mann–Kendall test · Hydroclimatic variability · Precipitation · Arno River

Introduction

Climate change is increasingly altering the global water cycle, with observable shifts in temperature, precipitation regimes, and hydrological responses documented across many regions of the world (IPCC 2021; Ahmed et al. 2026).

The Mediterranean region has been identified as a primary climate change hotspot, where warming rates exceed the global mean and hydroclimatic variability is amplified by the interaction of large-scale atmospheric circulation, complex topography, and land–sea contrasts (Giorgi 2006; Giorgi and Lionello 2008; Lionello and Scarascia 2018).

In such regions, relatively modest changes in mean climate can translate into substantial impacts on water resources, ecosystems, and socio-economic systems, especially where dense populations and intensive water use converge in climatically sensitive catchments (Cramer et al. 2018; IPCC 2022; Ahmed et al. 2026).

The Mediterranean climate is characterized by mild, wet winters and warm to hot, dry summers, with precipitation largely concentrated in the cool and transitional seasons and strong year-to-year variability superimposed on this pronounced seasonality (Xoplaki et al. 2004; Lionello et al. 2006). In many Mediterranean basins, a large fraction of annual runoff is generated during a relatively short wet season, while prolonged summer dry spells lead to low flows and recurrent hydrological droughts (Merheb et al. 2016; Cid et al. 2017). This combination of strong seasonality and high interannual variability makes Mediterranean river systems particularly sensitive to shifts in atmospheric circulation, changes in storm tracks, and warming-induced increases in evaporative demand (García-Ruiz et al. 2011; Vicente-Serrano et al. 2014; Barcikowska et al. 2020). Understanding how temperature, precipitation, and streamflow have co-evolved in recent decades at the basin scale is therefore essential for assessing ongoing climate impacts and informing adaptation strategies.

Italy, located in the central Mediterranean, exemplifies these climate-related challenges. Long instrumental records show coherent warming across the peninsula, with mean air temperature increases of approximately 1 °C over the period 1865–2003 (Brunetti et al. 2006), and continued warming projected for the coming decades under various emission scenarios (Spano et al. 2020). Precipitation trends, in contrast, are spatially heterogeneous and often weak or non-significant at regional scales, although tendencies toward increased variability and changes in the frequency of droughts and heavy precipitation events have been reported (Longobardi and Villani 2010; Philandras et al. 2011; Caloiero et al. 2018). Streamflow trends across Italian and broader Mediterranean catchments have yielded mixed results, with some basins exhibiting declining trends attributed to reduced precipitation and increased evapotranspiration, while others show no significant change (Stahl et al. 2010; Masseroni et al. 2021). These findings underscore the importance of basin-scale assessments that capture local responses to regional climate forcing.

Within this context, the Arno River basin in Tuscany is of particular scientific and societal interest. Covering approximately 8228 km², the basin drains a large portion of central Tuscany from the Apennine headwaters to the Ligurian Sea and encompasses major urban centers including Florence and Pisa (Diodato et al. 2021; Ghinassi et al. 2023). The Arno River has a long history of hydrological extremes,

including devastating floods, most notably the catastrophic 1966 Florence flood, and severe low-flow episodes that have periodically impacted infrastructure, economic activity, and cultural heritage (Caporali et al. 2005). The basin's hydrological regime is strongly seasonal, with high flows typically occurring from late autumn to early spring under cyclonic precipitation and saturated catchment conditions, and reduced discharges in summer under scarce rainfall and high evapotranspiration (Caporali et al. 2005; Castelli et al. 2009; Diodato et al. 2021). This makes the Arno River basin a representative case study for assessing hydroclimatic variability and change in Italian river systems.

Previous studies have addressed various aspects of Arno's hydroclimate across different temporal scales. A millennium-long storm erosivity reconstruction revealed the interplay between climate variability and erosive events (Diodato et al. 2021). Climate change effects on hydrological functionality and water-related ecosystem services have been evaluated in the upper Arno River basin using Soil and Water Assessment Tool (SWAT) modelling, indicating declining streamflow trends and negative impacts on provisioning and supporting services under future scenarios (El Jeitany et al. 2024b), while landscape metrics have been shown to significantly influence runoff and groundwater recharge predictions across the basin (El Jeitany et al. 2024a). Earlier work established the Arno River basin as representative of southern European climatic environments and documented expected changes in water balance seasonality and extremes (Burlando and Rosso 2002), with more recent projections indicating increasing temperatures and reductions in annual runoff by the 2050s under RCP4.5 and RCP8.5 scenarios (D'Oria et al. 2019). At broader scales, studies have examined precipitation and streamflow trends across Italian and Mediterranean catchments (Longobardi and Villani 2010; Merheb et al. 2016; Caloiero et al. 2018; Masseroni et al. 2021). However, relatively rare observational studies have focused specifically on the recent hydroclimatic behavior of the Arno River basin, jointly analyzing temperature, precipitation, and streamflow variability and trends over a consistent multi-decadal period.

Despite advances in regional climate and hydrological research, recent observational syntheses that jointly examine temperature, precipitation, and streamflow variability in the Arno River basin remain limited. Previous studies have addressed storm erosivity, hydrological modelling, projected climate-change impacts, and drought behavior, but fewer studies have evaluated recent observed co-variability among major hydroclimatic variables over a consistent multi-decadal period (Brunetti et al. 2006; Hertig and Trambly 2017; Diodato et al. 2021; Talpur et al. 2026). There is a need for updated observational analyses that explicitly include the most recent decades, during which

strong warming signals and several notable hydroclimatic anomalies have been observed in central Italy (Spano et al. 2020; IPCC 2022). Therefore, the present study is framed as a basin-scale observational assessment rather than a causal attribution or process-modelling. Its purpose is to quantify recent variability, monotonic trends, and statistical associations among temperature, precipitation, and streamflow, while acknowledging the limitations imposed by a 30-year record, station-network density, and single-gauge streamflow representation.

The objectives of this study are to (1) characterize the recent climatological patterns of temperature and precipitation across the Arno River basin; (2) quantify the interannual and seasonal variability of temperature, precipitation, and streamflow; (3) detect and quantify annual and seasonal monotonic trends using autocorrelation-aware non-parametric statistics; and (4) assess the observed co-variability among temperature, precipitation, and streamflow. The findings provide an updated observational baseline for water-resource monitoring and climate-adaptation planning in the basin.

Materials and methods

Study area

The Arno River basin is located in the Tuscany region of central Italy and drains westwards into the Ligurian Sea. The basin extends nearly between 42°40′–44°10′ N and 10°00′–12°00′ E and covers an area of about 8230 km², with elevations ranging from near sea level at the outlet to ~1625 above sea level (m a.s.l.) along the Apennines (Fig. 1a–c). The climate is Mediterranean, characterized by mild, wet winters and warm to hot, dry summers, modulated by Atlantic synoptic forcing and strong orographic effects (Lionello et al. 2006; Diodato et al. 2021). The hydrological regime of the Arno River is strongly seasonal with high flows usually occurring from late autumn to early spring in response to cyclonic precipitation and saturated catchment conditions, whereas low flows prevail during summer when precipitation is scarce and evapotranspiration demand is high. Land use is dominated by mixed agricultural and urban areas in the lowlands and forested and shrub-covered slopes in the upper basin (Diodato et al. 2021; El Jeitany et al. 2024b).

Hydro-meteorological data, quality control, and seasonal indexing

Daily precipitation and air temperature records (1994–2023) were obtained from the regional hydrological service of Tuscany (Servizio Idrologico Regionale della Toscana,

SIR). The station network comprises 18 precipitation stations and 7 temperature stations distributed across the basin. Streamflow was analyzed at the S. Giovanni alla Vena hydrometric station, located in the lower part of the basin. The digital elevation model was obtained from the Copernicus DEM GLO-30 dataset via the CREODIAS platform at 30-m resolution and was used for basin representation and mapping support (Fig. 1; Table 1).

Daily precipitation totals were aggregated to monthly, seasonal, and annual totals, while daily maximum and minimum temperature were used to calculate daily mean temperature and then aggregated to monthly, seasonal, and annual means. Monthly values were retained only when at least 25 daily observations were available, annual values when at least 350 daily observations were available, and seasonal values when at least 80 daily observations were available. Streamflow values (m³ s^{−1}) were converted to equivalent water depth (mm) using the contributing catchment area to facilitate comparison with precipitation totals. Seasons were defined as spring (MAM), summer (JJA), autumn (SON), and winter (DJF). Although the hydroclimate shows a broad wet–dry seasonal contrast, the four-season classification was retained because it separates spring and autumn transition seasons, which showed distinct precipitation-runoff behavior, and allows comparison with standard climatological seasons. Winter seasons were indexed by the year of January–February; for example, December 2010 was grouped with January and February 2011 as DJF 2011. Consequently, incomplete edge winters at the beginning and end of the record were excluded from seasonal contribution calculations. Annual streamflow records for 2003 and 2014 did not meet the ≥350-day threshold and were excluded from annual streamflow trend analysis. At seasonal scale, JJA 2003, DJF 2005, and JJA 2014 streamflow records did not meet the ≥80-day threshold and were excluded.

Station-level data completeness and homogeneity were assessed prior to basin aggregation. Per-station quality-control and homogeneity diagnostics are reported in (Supplementary Table S1). The table includes missing days, percentage missingness, longest data gap, number of valid annual and seasonal records, and annual-series homogeneity diagnostics based on Pettitt and Buishand tests. These tests were used as diagnostic checks for possible discontinuities; however, full metadata-based relative homogenization was beyond the scope of this study because station-history records on relocation, instrumentation, and observation-time changes were not examined in detail.

Basin averaging and spatial interpolation

Basin-averaged precipitation and temperature time series were first calculated using the arithmetic mean of available

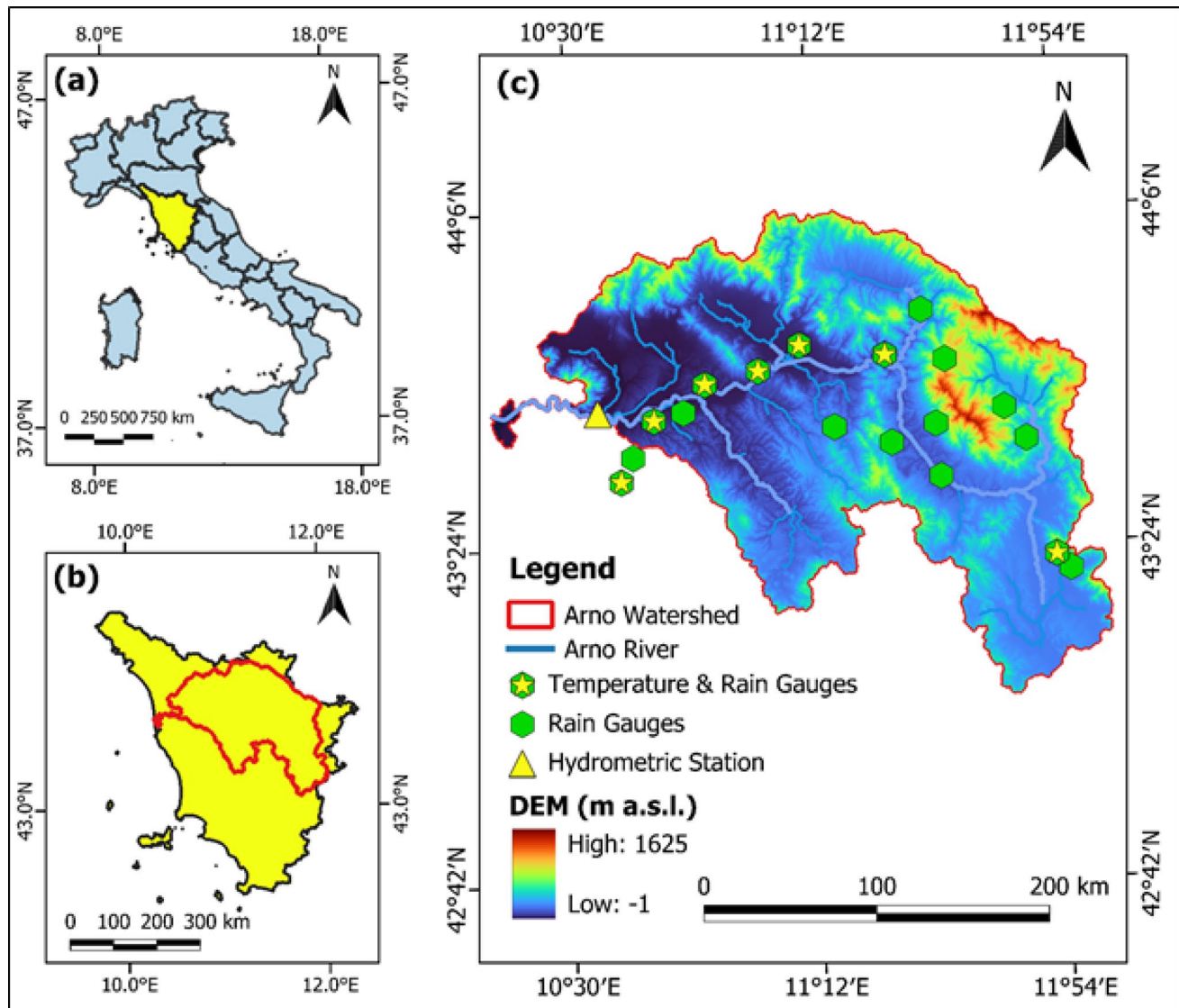


Fig. 1 Location of the study area: **a** Italy with the Tuscany region highlighted, **b** Tuscany region with the Arno River Basin boundary, and **c** Digital elevation model (DEM) of the Arno watershed with river

network, rain-gauge and temperature stations, and the S. Giovanni alla Vena hydrometric station

stations for each day, following previous basin-scale observational studies (Asfaw et al. 2018; Orke and Li 2021). Because the Arno River basin has strong topographic gradients, this choice was evaluated through a Thiessen-polygon sensitivity analysis. Thiessen polygons were generated from station coordinates in UTM Zone 32 N (EPSG:32632), clipped to the basin boundary, and used to calculate area-weighted daily precipitation and temperature series. Annual and seasonal arithmetic and Thiessen-weighted series were then compared using mean difference, median difference, percentage difference, and Pearson correlation.

Mean annual and seasonal precipitation maps were produced using inverse distance weighting (IDW) interpolation in QGIS v3.34. IDW is a deterministic interpolation method in which the estimated value at an unsampled location x_0

is calculated as a distance-weighted average of surrounding station observations (Shepard 1968; Lu and Wong 2008).

$$\hat{P}(x_0) = \frac{\sum_{i=1}^k w_i(x_0) P_i}{\sum_{i=1}^k w_i(x_0)}, \quad w_i(x_0) = d(x_0, x_i)^{-p} \quad (1)$$

where P_i is the precipitation observed at station i , $d(x_0, x_i)$ is the Euclidean distance between the prediction location x_0 and station i , p is the distance-decay power, and k is the number of neighboring stations included in the interpolation. IDW was implemented with $p = 2$ and $k = 8$ nearest stations using projected coordinates in UTM Zone 32 N (EPSG:32632), so that distances were calculated in meters.

Table 1 Precipitation, temperature, and streamflow gauging stations used in this study

S.No	Station number	Station name	Data type	Latitude (°N)	Longitude (°E)	Elevation
1	TOS01000601	Ortignano	(Precip)	43.682	11.752	346
2	TOS01000651	Salutio	(Precip)	43.615	11.813	315
3	TOS01000761	Castiglione Fiorentino	(Precip)	43.343	11.929	270
4	TOS01000881	Pian di Scò	(Precip)	43.65	11.555	395
5	TOS01000891	Il Palagio	(Precip)	43.613	11.427	315
6	TOS01001029	Dicomano	(Precip)	43.888	11.52	148
7	TOS01001041	Consuma	(Precip)	43.784	11.585	955
8	TOS01001129	Ferrone	(Precip)	43.648	11.264	150
9	TOS01001225	Case Passerini	(Precip), Temp)	43.82	11.168	33
10	TOS01001491	S. Miniato (Cimitero)	(Precip)	43.684	10.831	102
11	TOS01004571	Montevarchi	(Precip)	43.541	11.566	133.8
12	TOS01005131	Capannoli	(Precip)	43.592	10.684	28.61
13	TOS11000016	Terricciola	(Precip), Temp)	43.542	10.649	127
14	TOS11000024	Remole	(Precip, Temp)	43.796	11.413	230
15	TOS11000038	Ottavo	(Precip, Temp)	43.373	11.89	313
16	TOS11000046	Montopoli	(Precip, Temp)	43.668	10.746	29
17	TOS11000071	Cerreto Guidi	(Precip, Temp)	43.742	10.894	70
18	TOS11000076	Artimino	(Precip, Temp)	43.768	11.048	107
19	TOS01005191	S.Giovanni alla VV	River Discharge	43.685	10.585	7.63

All records are daily and cover 1994–2023. Elevation is reported in meters above sea level (m a.s.l.)

Leave-one-out cross-validation (LOOCV) was used to evaluate interpolation performance. LOOCV was executed in R using the gstat package (Pebesma 2004), whereby each station was sequentially withheld, the remaining stations were used to predict the value at the withheld location, and prediction errors were summarized using root mean square error (RMSE), RMSE as a percentage of the spatial mean, mean absolute error (MAE), bias, and coefficient of determination (R^2) (Supplementary Table S3). Ordinary kriging (OK) was also evaluated using the same LOOCV framework and neighborhood size ($k = 8$) to assess sensitivity to interpolation method. The comparison between IDW and OK is reported in Table S3. Given comparable accuracy and the greater parameter transparency of IDW, which avoids variogram model selection instability inherent to small station networks ($n = 18$), IDW was retained for all spatial mapping. The resulting gridded fields were clipped to the basin boundary and used to produce the precipitation maps shown in (Fig. 3).

Variability analysis

Hydroclimatic variability was quantified using the coefficient of variation (CV) and the standardized anomaly index (SAI), following common practice in regional climate studies (Asfaw et al. 2018; Orke and Li 2021).

Coefficient of variation (CV). The CV provides a dimensionless measure of relative variability and is defined as:

$$CV = \left(\frac{\sigma}{\mu} \right) \times 100 \quad (2)$$

Table 2 Classification of standardized anomaly index (SAI) values, following (McKee et al. 1993; Asfaw et al. 2018)

SAI value	Category
2.00 and above	Extremely wet
1.50 to 1.99	Very wet
1.00 to 1.49	Moderately wet
− 0.99 to 0.99	Near normal
− 1.49 to − 1.00	Moderately dry
− 1.99 to − 1.50	Severely dry
− 2.00 and less	Extremely dry

where σ is the standard deviation and μ is the long-term mean of the series, CV values were classified as indicating low ($CV < 20\%$), moderate ($20\text{--}30\%$), or high ($> 30\%$) variability (Orke and Li 2021).

Standardized anomaly index (SAI). To characterize relatively wet/dry years and seasons for precipitation and streamflow, standardized anomalies were calculated using the SAI (McKee et al. 1993; Asfaw et al. 2018).

$$SAI_t = \frac{X_t - \mu}{\sigma} \quad (3)$$

where X_t is the annual or seasonal value in year t and μ and σ are the corresponding long-term mean and standard deviation over 1994–2023. SAI values were classified using the categories in Table 2.

Trend analysis

Long-term monotonic trends in annual and seasonal basin-averaged temperature, precipitation and streamflow were assessed using a modified Mann–Kendall (MK) test that corrects for serial correlation in the data, combined with Sen's slope estimator to quantify trend magnitude (Mann 1945; Sen 1968; Kendall 1975; Hamed and Ramachandra Rao 1998; Yue and Wang 2004). The classical MK test is a rank-based nonparametric method widely used in hydro-climatology because it does not require normally distributed residuals and is relatively robust to outliers (Helsel and Hirsch 2002; Yue et al. 2002). The MK test statistic S is defined as:

$$S = \sum_{i=1}^{n-1} \sum_{j=i+1}^n \text{sgn}(X_j - X_i) \quad (4)$$

where n is the length of the series, X_i and X_j are observations at times i and j , and $\text{sgn}()$ is the sign function. Under the null hypothesis of no trend, S has mean zero and variance $\text{Var}(S)$ that depends on n and ties. The standardized test statistic Z is given by:

$$Z = \begin{cases} \frac{S - 1}{\sqrt{\text{Var}(S)}} & \text{if } S > 0 \\ 0 & \text{if } S = 0 \\ \frac{S + 1}{\sqrt{\text{Var}(S)}} & \text{if } S < 0 \end{cases} \quad (5)$$

and is approximately standard normal for $n \gtrsim 10$ (Helsel and Hirsch 2002). Because hydroclimatic time series often exhibit positive autocorrelation, which inflates the variance of S and may lead to over-estimation of trend significance, we applied the variance-corrected modified MK test proposed by (Hamed and Ramachandra Rao 1998; Yue and Wang 2004). In this approach, the effective sample size is reduced as a function of statistically significant positive autocorrelation in the ranks, and $\text{Var}(S)$ is adjusted accordingly, for estimating the effective sample size and corrected test statistics. Trend magnitudes were estimated using Sen's slope estimator (Sen 1968), which calculates the median of all pairwise slopes between data points by:

$$\beta = \text{median} \left(\frac{X_j - X_k}{j - k} \right), \quad j > k \quad (6)$$

The resulting slopes β are expressed per year $^{\circ}\text{C yr}^{-1}$ for temperature, mm yr^{-1} for precipitation and streamflow. Positive (negative) values of β indicate increasing (decreasing) trends. For each series, we report Kendall's tau (τ),

the modified MK Z statistic, the corresponding two-sided p value and Sen's slope. Statistical significance was evaluated at $p < 0.05$, 0.01 , and 0.001 . The 95% confidence intervals for Sen's slope were computed following the rank-based method of (Helsel and Hirsch 2002), in which the lower and upper bounds correspond to specific ranked pairwise slopes determined by the variance of S . The lag-1 autocorrelation coefficients, effective sample sizes, and variance inflation factors for each variable and season are reported in (Supplementary Table S4). To account for multiple comparisons across the 15 hypothesis tests ($5 \text{ periods} \times 3 \text{ variables}$), p values were also adjusted using the Benjamini–Hochberg false discovery rate (FDR) correction (Benjamini and Hochberg 1995). Both raw and adjusted p values are reported in Table 5.

Change-point analysis

To complement the monotonic trend analysis and test for abrupt shifts in the hydroclimatic series, change-point analysis was performed on annual basin-averaged temperature, precipitation, and streamflow using the Pettitt test (Pettitt 1979) and the Buishand range test (Buishand 1982). The Pettitt test is a nonparametric method that identifies the most probable single change-point location in a time series and tests whether the distributions before and after the change point differ significantly.

Correlation analysis

To examine the co-variability between basin-averaged temperature, precipitation, and streamflow, non-parametric Spearman rank correlation coefficients (ρ) were computed at both annual and seasonal scales (Spearman 1904; Helsel and Hirsch 2002). Spearman's ρ measures the strength and direction of a monotonic association between two variables based on the ranks of the data, and is less sensitive to outliers and non-normal distributions than the Pearson correlation. The coefficient is defined as:

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)} \quad (7)$$

where d_i is the difference between the ranks of corresponding values of the two variables and n is the number of paired observations.

Spearman's ρ captures monotonic associations but does not account for threshold-dependent hydrological processes such as saturation-excess runoff generation, infiltration capacity exceedance, or soil-moisture-mediated nonlinearities, which may introduce more complex

precipitation–streamflow relationships not fully captured by a rank-based measure.

Results

Climate characteristics of the study area

Station-level quality-control diagnostics showed generally high data completeness, with missing values ranging from 0.16 to 8.07% for precipitation stations and 0.69–3.17% for temperature stations. No precipitation station showed significant Pettitt or Buishand discontinuities at $p < 0.05$. In contrast, temperature stations commonly showed significant change-point diagnostics, mostly between 2008 and 2013, consistent with the basin-scale warming shift identified in the annual T_{avg} series. These results were therefore treated as diagnostic evidence of temporal structure rather than isolated proof of station artifacts (Supplementary Table S1). Over the period 1994–2023, the Arno River basin exhibits a pronounced seasonal cycle in both temperature and precipitation (Fig. 2). Mean monthly air temperature shows a typical Mediterranean pattern, increasing from a minimum basin-averaged value of 6.4 °C in January to a maximum of 24.4 °C in August, and then declining towards early winter. Mean monthly maximum temperature T_{max} rises from 10.31 °C in January to 31.18 °C in July and August, while minimum temperature T_{min} increases from 2.54 °C in January to 17.45 and 17.58 °C in July and August, respectively. Seasonally, mean average temperature T_{avg} is lowest in winter 7.00 °C, increases through spring 13.66 °C, peaks in summer 23.46 °C, and remains relatively mild in autumn 15.69 °C. Basin-averaged precipitation also displays a strong intra-annual variability, with a pronounced autumn–winter maximum and a dry summer (Fig. 2). The

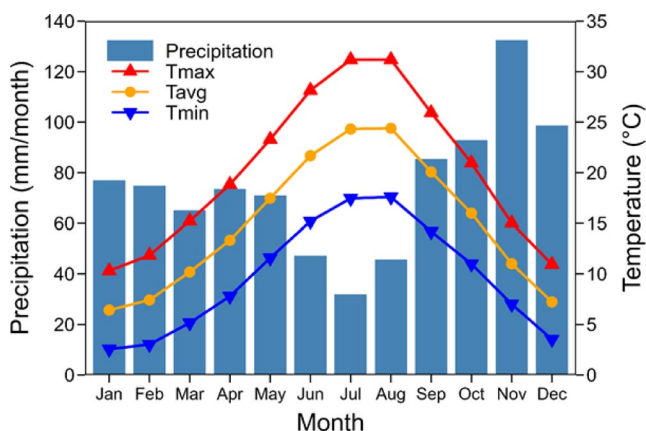


Fig. 2 Monthly climatology of basin-averaged precipitation and temperature for 1994–2023. Bars show mean monthly precipitation totals on the left axis, and lines show maximum temperature (T_{max}), mean temperature (T_{avg}), and minimum temperature (T_{min}) on the right axis

wettest month is November, with a mean total of 132.5 mm, followed by December 98.7 mm, whereas July is the driest month with only 32 mm. The long-term mean annual precipitation is approximately 895.8 mm, of which about 34.3% falls in autumn, 28.3% in winter, 23.3% in spring, and only 14.1% in summer. This distribution highlights the concentration of precipitation in the cool and transitional seasons and the relative dryness of the summer months in the Arno River basin. The four-season partitioning also highlights the distinct role of transition seasons. Autumn contributed the largest share of annual precipitation, whereas winter contributed the largest share of annual streamflow, indicating that precipitation and runoff seasonality are related but not identical.

Spatial distribution of annual and seasonal precipitation

The spatial distribution of precipitation across the Arno River basin on annual and seasonal scales is presented in Fig. 3. Annual precipitation (Fig. 3a) exhibits considerable spatial variability, ranging from approximately 720–830 mm in the central and southwestern lowlands to more than 1160 mm in the northeastern headwater areas along the Apennine ridge. This pattern indicates a pronounced orographic gradient, with precipitation increasing progressively from the lower Arno valley towards the elevated mountainous sectors. Seasonal maps (Fig. 3b–e) demonstrate that this orographic influence persists throughout the year, though with varying intensity across seasons. In spring (Fig. 3b), precipitation totals range between 175 and 315 mm, with maximum values concentrated in the northeastern uplands and minimum values over the central plains. Summer (Fig. 3c) represents the driest season, with precipitation totals predominantly between 90 and 170 mm; even the orographic enhancement over the mountainous areas remains modest during this period. Autumn (Fig. 3d) exhibits a marked increase in precipitation, with totals ranging from 240 to 410 mm. The highest values are again concentrated along the Apennine crest, while lower amounts characterize the central and western lowlands. Winter (Fig. 3e) displays relatively high totals between 180 and 370 mm, with a spatial pattern broadly similar to autumn, though slightly weaker in magnitude.

This spatial variation in precipitation distribution across the watershed is mainly attributed to the orographic effect of the Apennine Mountains, which enhance precipitation on their windward slopes. The pattern reveals a consistent northeast-to-southwest gradient, with higher precipitation in elevated areas and lower values in the coastal and valley lowlands throughout all seasons.

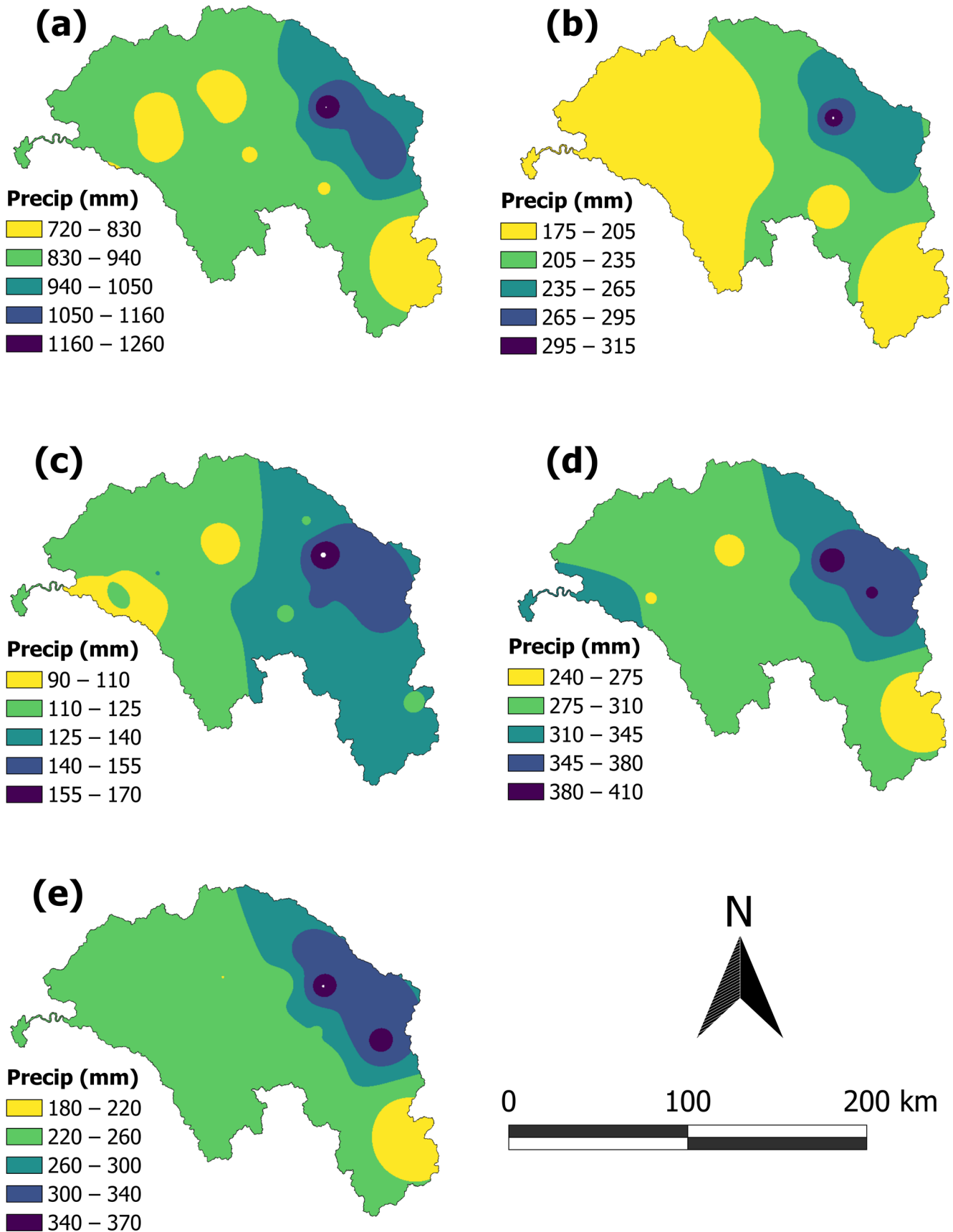


Fig. 3 Spatial distribution of mean precipitation (mm) across the Arno River basin for 1994–2023, interpolated using inverse distance weighting (IDW; $p = 2$, $k = 8$): **a** annual, **b** spring (MAM), **c** summer (JJA), **d** autumn (SON), and **e** winter (DJF)

Table 3 Mean seasonal contribution of precipitation and streamflow to annual totals

Season	Precipitation mean (%)	Precipitation SD (%)	Streamflow mean (%)	Streamflow SD (%)
Spring	23.32	5.08	32.24	8.17
Summer	14.09	5.03	6.73	2.76
Autumn	34.29	8.17	17.63	11.93
Winter	28.31	7.49	43.40	13.07

Percentages were calculated for each valid year and then averaged across years. Winter (DJF) was indexed by the year of January–February; incomplete edge winters were excluded

The Thiessen-polygon sensitivity analysis showed that arithmetic and Thiessen-weighted basin averages were highly consistent (Supplementary Table S2). Annual precipitation from the Thiessen-weighted series averaged 877.4 mm compared with 895.8 mm from the arithmetic method, corresponding to a median relative difference of approximately -2.0% and a Pearson correlation of $r = 0.996$. Thiessen-weighted annual T_{avg} was on average $0.19\text{ }^{\circ}\text{C}$ lower than the arithmetic mean, also with $r = 0.996$. Seasonal comparisons showed similarly small differences. These results indicate that the main trend and correlation conclusions were not sensitive to the basin-averaging method.

The LOOCV validation of IDW interpolation ($p = 2$, $k = 8$) indicated moderate interpolation skill across annual and seasonal precipitation fields (Supplementary Table S3). Annual IDW RMSE was 105.2 mm, corresponding to 11.8% of the spatial mean, with $MAE = 83.6\text{ mm}$, $bias = 2.2\text{ mm}$, and $R^2 = 0.44$. Seasonal relative RMSE ranged from 10.9% in autumn to 14.1% in winter. Ordinary kriging produced comparable but not consistently better performance: it slightly improved RMSE in spring and autumn, while IDW performed better for annual, summer, and winter fields. Given this comparable performance and the greater transparency of the fixed IDW parameterization under a relatively small station network, IDW was retained for spatial mapping. The resulting precipitation maps are therefore interpreted as descriptive station-based surfaces rather than elevation-corrected precipitation reconstructions.

Seasonal contribution to annual precipitation and streamflow

Seasonal contributions of precipitation and streamflow to the annual water balance exhibit marked differences in the Arno River watershed Table 3. On average, autumn accounts for the largest share of annual precipitation 34.29% , followed by winter 28.31% , spring 23.32% , and summer with only 14.09% . The interannual variability of these seasonal fractions is moderate for spring and summer $SD \approx 5\%$, but

Table 4 Interannual coefficients of variation (CV, %) of seasonal and annual basin-averaged mean temperature (T_{avg}), maximum temperature (T_{max}), minimum temperature (T_{min}), precipitation, and streamflow for 1994–2023

Season	CV T_{avg} (%)	CV T_{max} (%)	CV T_{min} (%)	CV Precipitation (%)	CV Streamflow (%)
Spring	5.7	5.2	9.3	29.5	46.2
Summer	4.8	4.7	5.4	36	36.3
Autumn	6.2	5.3	9.5	29.8	65.8
Winter	15.7	10.4	39.3	31.3	48.3
Annual	4.5	3.9	6.6	18.2	33.4

CV values were calculated using records that met the completeness thresholds described in Sect. 2.2

considerably larger in autumn 8.17% and winter 7.49% . In contrast, streamflow is strongly concentrated in the cold season. Winter contributes on average 43.40% of the annual runoff, followed by spring 32.24% , whereas autumn and summer account for only 17.63% and 6.73% , respectively. The standard deviation of seasonal runoff fractions is highest in winter 13.07% and autumn 11.93% and substantially lower in spring 8.17% and summer 2.76% . These patterns indicate substantial year-to-year variability in the seasonal distribution of discharge, particularly during the wettest months.

Variability analysis

Temperature variability

Annual temperature variability in the Arno River basin was assessed using (CV) and (SAI) for the period 1994–2023. The annual CV of T_{min} 6.6% was higher than those of T_{avg} 4.5% and T_{max} 3.9% , indicating that interannual variability is modest for mean and maximum temperature but substantially higher for minimum temperature. Seasonally, winter exhibited the strongest variability, with CV values of 15.7% , 10.4% , and 39.3% for T_{avg} , T_{max} , and T_{min} , respectively, whereas summer displayed the lowest variability CV of 4.8% , 4.7% , and 5.4% for T_{avg} , T_{max} , and T_{min} . Spring and autumn showed intermediate CVs, with T_{min} systematically more variable than T_{max} and T_{avg} in all seasons Table 4.

Figure 4 shows that annual temperature anomalies for all three temperatures were predominantly negative or near zero from 1994 to about 2013, followed by a sequence of mostly positive anomalies from 2014 onward, with markedly warm years concentrated toward the end of the record. The strongest negative anomalies varied: T_{max} in 2010 $-1.56\text{ }^{\circ}\text{C}$, T_{avg} in 2005 $-1.31\text{ }^{\circ}\text{C}$ and T_{min} in 1995 $-1.33\text{ }^{\circ}\text{C}$. In contrast, 2022 and 2023 recorded the strongest positive anomalies, reaching up to $+1.94\text{ }^{\circ}\text{C}$ for T_{max} , $+1.54\text{ }^{\circ}\text{C}$ for T_{avg} , and $+1.37\text{ }^{\circ}\text{C}$ for T_{min} . Linear regression fitted to the anomaly series indicates coherent warming across all temperatures,

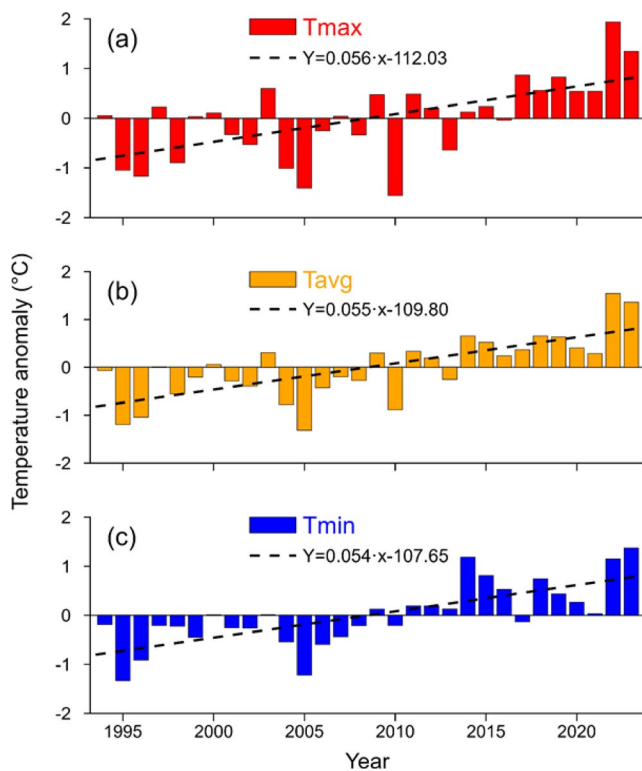


Fig. 4 Annual temperature anomalies for **a** maximum temperature ($T_{\max}$), **b** mean temperature (T_{avg}), and **c** minimum temperature ($T_{\min}$). Bars show anomalies in $^{\circ}\text{C}$ relative to the 1994–2023 mean, and dashed lines represent linear regression trends with equations shown in each panel

with trends of $0.056\text{ }^{\circ}\text{C yr}^{-1}$ for $T_{\max}$ and 0.055 and $0.054\text{ }^{\circ}\text{C yr}^{-1}$ for both T_{avg} and $T_{\min}$ (Fig. 4a–c). Taken together, the relatively low annual CV, the strong seasonal contrast in variability especially the high winter and $T_{\min}$ variability and predominance of positive anomalies after the mid-2010s is consistent with the warming signal quantified by the modified Mann–Kendall and Sen’s slope analyses over the past three decades. Notably, the CV of streamflow substantially exceeds that of precipitation in all seasons such as autumn: CV Streamflow 65.8% vs. CV Precipitation 29.8%; Table 4, indicating that catchment processes amplify precipitation variability into even greater streamflow variability. This amplification reflects the role of antecedent soil moisture conditions, groundwater storage dynamics, and threshold-dependent runoff generation mechanisms (Gallart et al. 2002; Merheb et al. 2016).

Precipitation variability

Annual precipitation in the basin shows moderate interannual variability, with a CV of 18.2%. Seasonally, summer and winter exhibited the highest CV of 36% and 31.3%, respectively, while spring and autumn showed slightly lower CV values of 29.5% and 29.8%, respectively Table 4.

This pattern reflects the sensitivity of cold-season and convective summer precipitation to synoptic-scale circulation anomalies over the central Mediterranean.

The annual SAI series (Fig. 5a) reveals alternating wet and dry years with no pronounced long-term trend of the 30 years analyzed, with 16 years exhibiting negative and 14 positive SAI values, indicating a slight predominance of drier years. The strongest positive anomaly occurred in 2010: $\text{SAI} = +2.55$, followed by 2013 and 2014: $\text{SAI} = +1.48$, $+1.43$. Conversely, the most pronounced negative anomalies were recorded in 2011: $\text{SAI} = -1.76$. Seasonal patterns exhibit distinct characteristics across seasons. Spring recorded its driest conditions in 2011 and 2017: $\text{SAI} \approx -1.42$, while the wettest springs occurred in 2013 and 2018: $\text{SAI} \approx +2.61$ (Fig. 5b). Summer showed the strongest negative anomalies in 2017: $\text{SAI} = -1.85$, with the wettest summers in 2002: $\text{SAI} = +2.40$ (Fig. 5c). Autumn displayed its most pronounced dry anomaly in 2011: $\text{SAI} = -1.94$, whereas 2019: $\text{SAI} = +1.87$ was the wettest autumn (Fig. 5d). Winter experienced its driest conditions in 2012: $\text{SAI} = -1.53$, and the wettest winters in 2021: $\text{SAI} = +2.33$ (Fig. 5e).

Notably, 2011 emerged as a consistently dry year across multiple seasons, with the strongest negative annual and the most pronounced autumn deficit. Similarly, 2017 recorded severe dry anomalies in spring, summer, and winter. In contrast, 2010 stands out as an exceptionally wet year, with the highest annual SAI driven primarily by abundant winter precipitation. The absence of a systematic trend in the SAI series, combined with moderate to high seasonal CVs, suggests that precipitation variability in the Arno River basin is characterized by substantial year-to-year fluctuations rather than a progressive shift toward wetter or drier conditions.

Streamflow variability

Annual and seasonal streamflow variability at the S. Giovanni alla Vena gauging station was assessed using CV and the SAI of streamflow (Table 4; Fig. 6). The annual CV of streamflow is 33.4%, indicating pronounced interannual variability, considerably higher than that of precipitation 18.2%. Seasonal CVs are even larger, ranging from 36.3% in summer to 65.8% in autumn, with spring and winter also highly variable 46.2% and 48.3%, respectively. Streamflow variability exceeded precipitation variability in all seasons, most notably in autumn, where streamflow CV reached 65.8% compared with 29.8% for precipitation. This amplification is consistent with catchment-memory effects, whereby antecedent soil moisture, groundwater storage, and threshold-dependent runoff generation can either dampen or amplify the streamflow response to seasonal precipitation. In autumn, early rainfall may first replenish summer

Fig. 5 Standardized anomaly index (SAI) of basin-averaged precipitation for the period 1994–2023: **a** Annual scale and **(b–e)** seasonal scales for **b** Spring (MAM), **c** Summer (JJA), **d** Autumn (SON) and **e** Winter (DJF)

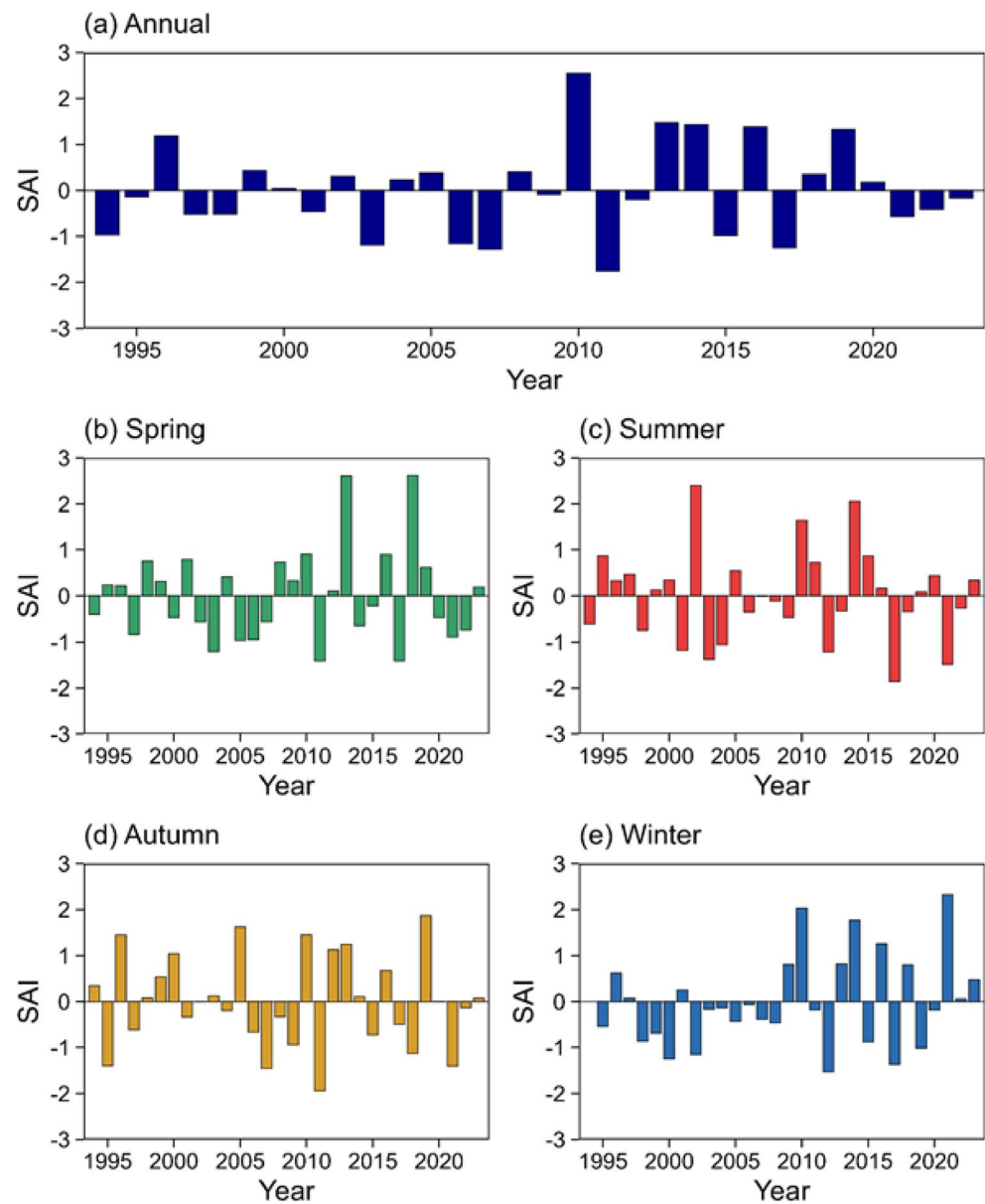
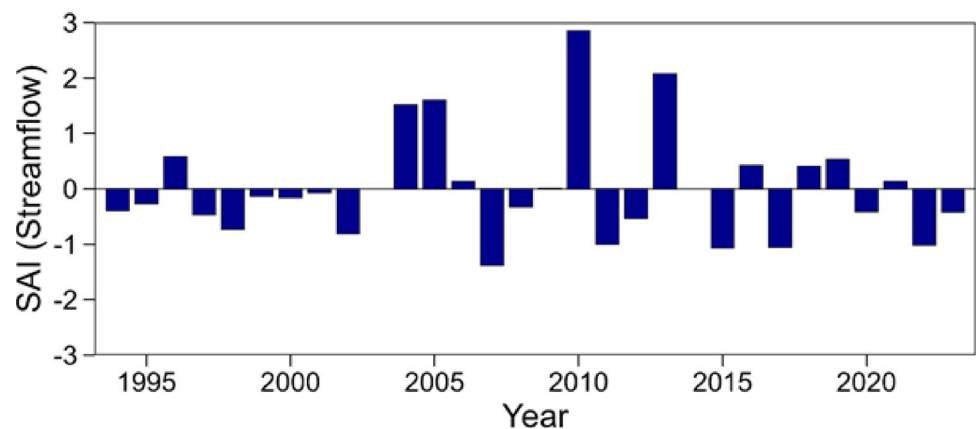


Fig. 6 Annual Standardized Anomaly Index (SAI) of streamflow at the S. Giovanni alla Vena gauging station for 1994–2023. Annual streamflow years not meeting the ≥ 350 -day completeness threshold were excluded



soil-moisture deficits before producing runoff, whereas rainfall following already wet antecedent conditions may generate disproportionately large streamflow responses.

The annual streamflow SAI series (Fig. 6) shows alternating high and low flow years without a clear long-term trend. Of the 28 years with sufficient data coverage, 17 exhibit negative SAI values and 11 positive values, indicating a predominance of below-average runoff conditions. The largest positive anomaly occurs in 2010 $\text{SAI}=+2.85$, followed by 2013 $+2.08$ and 2005 $+1.60$ corresponding to very wet years for the basin. The most pronounced negative anomaly is recorded in 2007 $\text{SAI}=-1.38$. Overall, the combination of high seasonal CVs and frequent positive and negative annual SAI values indicates a highly variable streamflow regime dominated by episodic high-flow years interspersed with numerous low-flow years, rather than a monotonic shift toward either wetter or drier hydrological conditions.

Trend analysis

Temperature trends

Trends in mean temperature were analyzed using the modified Mann–Kendall test and Sen's slope with 95% confidence intervals for the period 1994–2023 (Fig. 7; Table 5). Statistically significant warming was detected in summer and autumn, with both results significant Benjamini–Hochberg correction for multiple testing. Summer mean temperature increased at $0.066\text{ }^{\circ}\text{C yr}^{-1}$ $\tau=0.416$, $p=0.001$, $\text{pBH}=0.010$; 95% CI: $[0.036, 0.113]\text{ }^{\circ}\text{C yr}^{-1}$, and autumn at $0.073\text{ }^{\circ}\text{C yr}^{-1}$ $\tau=0.485$, $p<0.001$, $\text{pBH}=0.003$; 95% CI: $[0.042, 0.110]\text{ }^{\circ}\text{C yr}^{-1}$ (Fig. 7c and d; Table 5). Autumn thus displays the strongest seasonal trend, equivalent to roughly $2.2\text{ }^{\circ}\text{C}$ over the 30-year period. The annual mean temperature trend was nominally significant $0.052\text{ }^{\circ}\text{C yr}^{-1}$; $\tau=0.531$, $p=0.021$ but did not remain significant after BH correction $\text{pBH}=0.106$, reflecting substantial positive autocorrelation in the annual series lag-1 $\text{ACF}=0.45$, effective sample size $n^*\approx 9.5$, variance inflation factor $=3.17$; Table S4. Spring exhibited a weak non-significant trend $0.027\text{ }^{\circ}\text{C yr}^{-1}$; $\tau=0.195$, $p=0.134$, and winter showed a positive but non-significant trend $0.051\text{ }^{\circ}\text{C yr}^{-1}$; $\tau=0.236$, $p=0.075$. The confidence intervals for spring and winter trends span zero, indicating the absence of detectable warming in these seasons.

The warming signal was further supported by change-point analysis. The Pettitt test identified a significant change point in annual mean temperature at 2010, $p<0.001$, also supported by the Buishand range test $p=0.001$. No significant change points were detected for annual precipitation Pettitt $p=1.0$; Buishand $p=0.76$ or streamflow Pettitt

$p=1.0$; Buishand $p=0.68$. This change point is consistent with the concentration of positive temperature anomalies observed since the mid-2010s (Fig. 4) and with the coherent station-level change points detected across all seven temperature stations (Table S1).

Precipitation trends

No statistically significant monotonic precipitation trends were detected at annual or seasonal scales (Fig. 8; Table 5). Annual precipitation showed a small positive Sen's slope of $+1.113\text{ mm yr}^{-1}$, but the confidence interval included zero and the adjusted p value was non-significant $\text{pBH}=0.895$. Seasonal precipitation slopes were also non-significant, indicating no detectable monotonic precipitation trend within the available 30-year record. It should be noted that the absence of significant trends does not constitute evidence of precipitation stability; with $n=30$ and seasonal CVs of 29–36%, the modified MK test has limited statistical power to detect small-to-moderate monotonic trends (Yue et al. 2002).

Streamflow trends

No statistically significant monotonic streamflow trends were detected at annual or seasonal scales (Fig. 9; Table 5). Annual streamflow showed a small negative Sen's slope of -0.189 mm yr^{-1} , but the confidence interval included zero and the adjusted p value was non-significant $\text{pBH}=0.920$. These results indicate absence of a detectable monotonic trend in the single-gauge streamflow record, not evidence of hydrological stability.

Change points

Change-point diagnostics showed a significant annual Tavg shift around 2010, supported by Pettitt ($p<0.001$) and Buishand ($p=0.001$) tests (Supplementary Table S5). In contrast, annual precipitation and streamflow showed no significant change points. This supports the interpretation of a step-like warming shift around 2010 but does not demonstrate acceleration.

Spearman's rank correlation analysis

Seasonal and annual relationships among basin-averaged temperature, precipitation, and streamflow were evaluated using Spearman's rank correlation coefficient (ρ) Table 6. Precipitation and streamflow were positively correlated across all seasons, with Spearman's ρ ranging from 0.51 in spring to 0.93 in autumn. These correlations remained significant after Benjamini–Hochberg correction, indicating

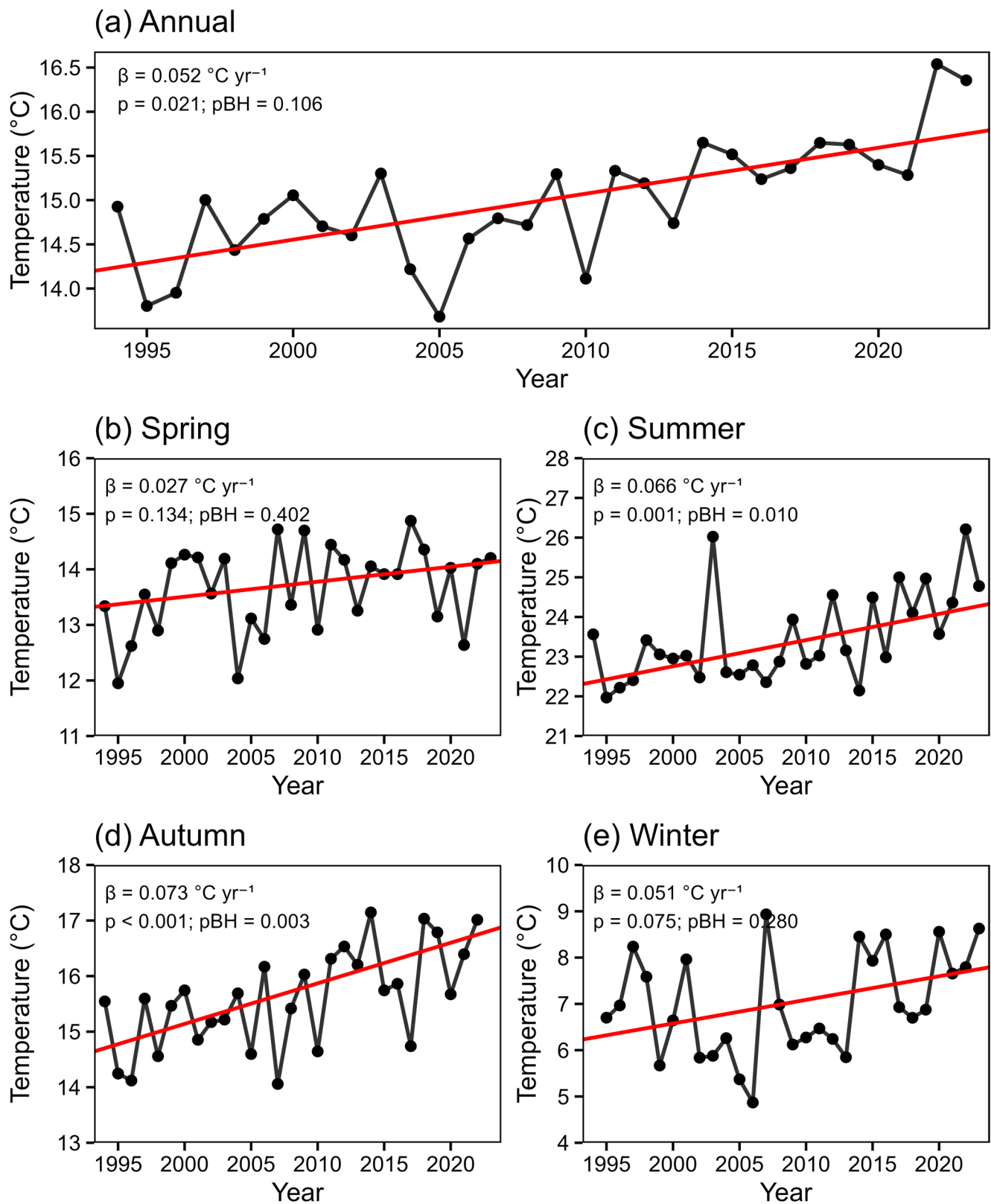


Fig. 7 Annual and seasonal trends in basin-averaged mean temperature (T_{avg}) for 1994–2023: **a** annual, **b** spring (MAM), **c** summer (JJA), **d** autumn (SON), and **e** winter (DJF). Red lines represent Sen's slope

estimates. Panel annotations show Sen's slope (β , $^\circ\text{C yr}^{-1}$), unadjusted p value, and Benjamini–Hochberg adjusted p value ($p\text{BH}$)

Table 5 Modified Mann–Kendall trend statistics and Sen's slope estimates with 95% confidence intervals for basin-averaged mean temperature (Tavg), precipitation (Precip), and streamflow in the Arno River basin, 1994–2023

Variable	Season	Kendall's τ	p value	pBH	Sen's slope	95% CI lower	95% CI upper	Sig
Tavg	Spring	0.195	0.134	0.402	0.027	− 0.007	0.065	ns
Tavg	Summer	0.416	0.001	0.01	0.066	0.036	0.113	**
Tavg	Autumn	0.485	<0.001	0.003	0.073	0.042	0.11	**
Tavg	Winter	0.236	0.075	0.28	0.051	− 0.004	0.099	ns
Tavg	Annual	0.531	0.021	0.106	0.052	0.011	0.1	ns(BH)
Precip	Spring	− 0.007	0.972	0.972	− 0.017	− 2.404	2.531	ns
Precip	Summer	− 0.039	0.775	0.895	− 0.329	− 2.172	1.6	ns
Precip	Autumn	− 0.071	0.61	0.895	− 1.092	− 6.108	3.927	ns
Precip	Winter	0.163	0.223	0.522	2.54	− 1.343	6.389	ns
Precip	Annual	0.044	0.748	0.895	1.113	− 5.943	9.453	ns
Streamflow	Spring	− 0.076	0.637	0.895	− 0.463	− 2.474	1.835	ns
Streamflow	Summer	− 0.159	0.244	0.522	− 0.198	− 0.501	0.156	ns
Streamflow	Autumn	− 0.076	0.568	0.895	− 0.349	− 1.818	0.756	ns
Streamflow	Winter	0.058	0.678	0.895	0.774	− 1.793	4.954	ns
Streamflow	Annual	− 0.026	0.859	0.92	− 0.189	− 3.782	3.328	ns

p is the unadjusted modified Mann–Kendall p value, and pBH is the Benjamini–Hochberg adjusted p value. Significance codes in the final column are based on pBH: * pBH<0.05; ** pBH<0.01; *** pBH<0.001; ns=not significant

strong seasonal co-variability between precipitation and streamflow. The annual precipitation–streamflow correlation was also positive and significant $\rho = 0.718$, pBH<0.001. Summer Tavg was negatively correlated with precipitation $\rho = -0.528$, pBH=0.008 and streamflow $\rho = -0.533$, pBH=0.009, indicating warm-season hydroclimatic sensitivity. However, these correlations do not identify a single mechanism because summer streamflow can also be affected by antecedent storage, abstractions, reservoir operation, and seasonal rainfall deficits.

Discussion

Regional hydroclimatic characteristics

The hydroclimatic characteristics of the Arno River basin documented in this study are consistent with the broader Mediterranean climate regime, characterized by warm, dry summers and mild, wet winters (Lionello et al. 2006). The observed seasonal precipitation distribution, with autumn contributing 34.29% and winter 28.31% of annual totals while summer accounts for only 14.09%, Table 3, indicates a strong cool-season concentration and a pronounced summer minimum, consistent with Mediterranean precipitation seasonality (Xoplaki et al. 2004; Lionello et al. 2006). This pronounced seasonality reflects the influence of large-scale circulation during the wet season, with variability linked to synoptic dynamics and broader circulation modes affecting Mediterranean precipitation (Xoplaki et al. 2004).

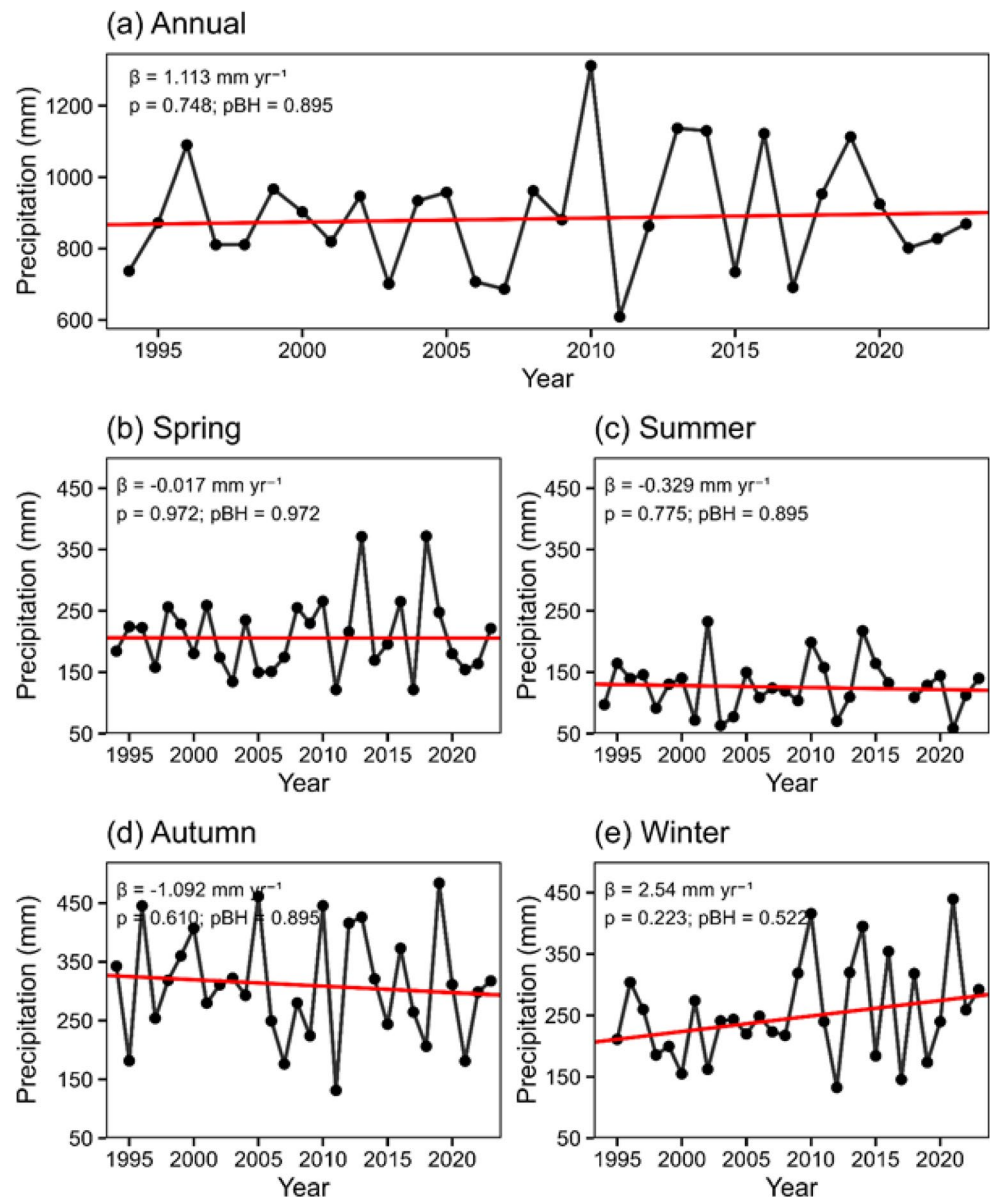
The spatial distribution of precipitation across the Arno River basin reveals a strong orographic gradient, with annual totals ranging from approximately 720 mm in the

central lowlands to over 1,160 mm along the Apennine ridge (Fig. 3a). Such topographic modulation of precipitation is a well-established feature of mountainous terrain and is documented for central Italy, where Apennine orography contributes to the spatial variability of rainfall (Daly et al. 1994; Silvestri et al. 2022). The persistence of this gradient across all seasons, though with reduced intensity during summer, underscores the fundamental role of orography in shaping the basin's water resources.

Temperature trends

The statistically significant warming detected in summer +0.066 °C yr^{−1} and autumn +0.073 °C yr^{−1} in the Arno River basin is consistent with assessments for the Mediterranean region, where surface temperature has exceeded +1.5 °C above pre-industrial levels and regional warming has outpaced the global average since the 1980s (Aruffo and Carlo 2019; IPCC 2022). The annual warming trend (+0.052 °C yr^{−1}), while nominally significant, did not remain significant after Benjamini–Hochberg correction for multiple testing (pBH=0.106), largely due to substantial positive autocorrelation in the annual series that reduced the effective sample size from 30 to approximately 9.5 years. Nevertheless, the warming signal is independently confirmed by the Pettitt change-point test (2010; $p < 0.001$) and by the coherent change points detected across all seven temperature stations Table S1, providing converging evidence of a real climatic shift. This 30-year analysis captures recent climate signals and aligns with the WMO standard climate normal period, though it may not fully resolve multi-decadal oscillations.

Fig. 8 Annual and seasonal trends in basin-averaged precipitation for 1994–2023: **a** annual, **b** spring (MAM), **c** summer (JJA), **d** autumn (SON), and **e** winter (DJF). Red lines represent Sen's slope estimates. Panel annotations show Sen's slope (β , mm yr⁻¹), unadjusted p value, and Benjamini–Hochberg adjusted p value (pBH)



The concentration of warming in summer and autumn has implications for atmospheric evaporative demand. Several studies in the Mediterranean region have documented significant increases in potential evapotranspiration driven by rising temperatures, even in basins where precipitation has remained stable (Chaouche et al. 2010; Vicente-Serrano et al. 2014; Dakhlaoui et al. 2020). This thermal pathway in which warming-driven increases in evaporative demand reduce water availability independently of precipitation trends has been identified as an emerging risk for Mediterranean water resources (García-Ruiz et al. 2011; IPCC 2022). The significant summer temperature–streamflow correlation observed in the present study ($\rho = -0.53$) can be consistent with this mechanism, although direct quantification of evapotranspiration trends would require dedicated PET

estimation and land-surface diagnostics beyond the scope of this observational synthesis.

Precipitation variability and absence of significant trends

Despite the robust warming signal, annual and seasonal precipitation in the Arno River basin showed no statistically significant trends over 1994–2023, exhibiting instead high interannual variability $CV = 18.2\%$ annually, up to 36% in summer Table 4. This outcome is consistent with Mediterranean-scale evidence that trends are spatially heterogeneous and often weak or non-significant at basin/regional scales, even where temperature trends are robust (Longobardi and Villani 2010; Philandras et al. 2011; Caloiero et al. 2019). At

Fig. 9 Annual and seasonal trends in streamflow at the S. Giovanni alla Vena gauging station for 1994–2023: **a** annual, **b** spring (MAM), **c** summer (JJA), **d** autumn (SON), and **e** winter (DJF). Red lines represent Sen's slope estimates. Panel annotations show Sen's slope (β , mm yr^{-1}), unadjusted p value, and Benjamini–Hochberg adjusted p value (pBH). Years or seasons not meeting the completeness thresholds described in Sect. 2.2 were excluded

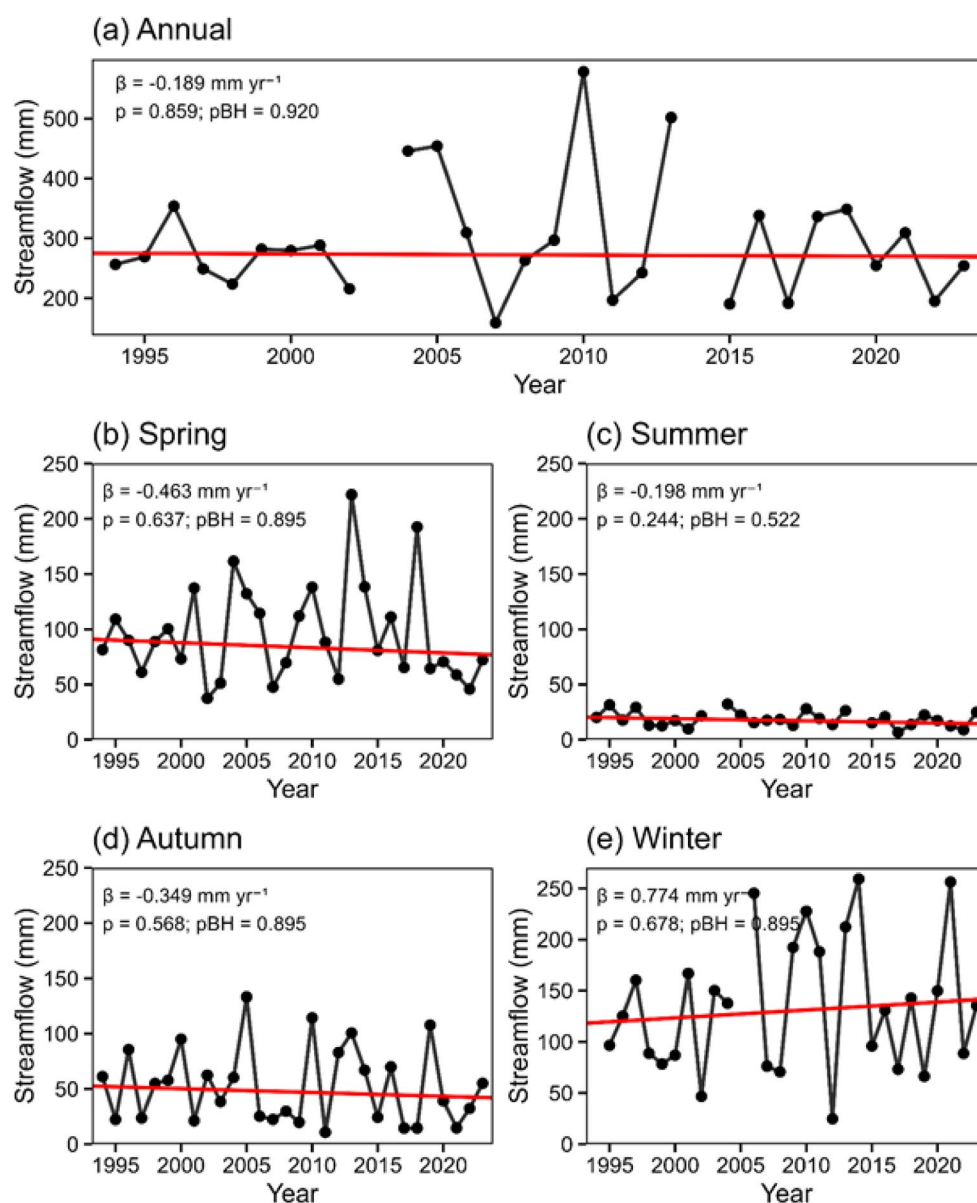


Table 6 Seasonal and annual Spearman rank correlations among basin-averaged mean temperature (T_{avg}), precipitation, and streamflow for 1994–2023

Season	Temp & Precip correlation	Temp & Precip p value	Temp & Precip BH-adjusted p value	Temp & Streamflow correlation	Temp & Streamflow p value	Temp & Streamflow BH-adjusted p value	Precip & Streamflow correlation	Precip & Streamflow p value	Precip & Streamflow BH-adjusted p value
Spring	− 0.186	0.326	0.412	− 0.23	0.221	0.368	0.51	0.004	0.009
Summer	− 0.528	0.003	0.008	− 0.533	0.003	0.009	0.551	0.002	0.008
Autumn	− 0.041	0.831	0.959	− 0.025	0.895	0.959	0.93	0	0
Winter	0.208	0.279	0.412	− 0.001	0.996	0.996	0.807	0	0
Annual	− 0.184	0.329	0.412	− 0.404	0.033	0.062	0.718	0	0

Values include Spearman's ρ , unadjusted p values, and Benjamini–Hochberg adjusted p values. Correlations were calculated using paired records that met the completeness thresholds described in Sect. 2.2

a sub-regional scale, similarly, no trend in annual precipitation was reported for a western French Mediterranean study area over 1970–2006, despite clear increases in temperature and potential evapotranspiration (Chaouche et al. 2010).

The alternating sequence of wet and dry years, with 2010 emerging as very wet $SAI=+2.55$ and 2011 as severely dry $SAI=-1.76$ (Fig. 5a), illustrates the episodic nature of Mediterranean precipitation variability. Such year-to-year swings are strongly influenced by large-scale atmospheric circulation, including variability associated with the North Atlantic Oscillation (NAO), which modulates precipitation over large parts of Europe and the Mediterranean basin (Hurrell and Van Loon 1997; Philandras et al. 2011). The absence of a systematic trend, combined with high seasonal variability, suggests that future water resource planning in the Arno River basin should prioritize adaptation to interannual variability rather than monotonic change.

Streamflow dynamics and precipitation-runoff relationships

The streamflow regime of the Arno River exhibits pronounced seasonality, with winter and spring together contributing 76% of annual runoff despite receiving only 52% of annual precipitation Table 3. This seasonal redistribution is consistent with Mediterranean catchment behavior, where cool-season runoff is supported by lower evaporative demand and catchment wetting, while summer drought conditions promote soil moisture depletion and reduced discharge (Gallart et al. 2002; Merheb et al. 2016). During summer, high evaporative losses and limited rainfall typically translate into strongly reduced flows, a characteristic of Mediterranean-climate river regimes (Cid et al. 2017; Lupon et al. 2018).

The strong positive correlation between precipitation and streamflow ($\rho=0.72$ annually, reaching $\rho=0.93$ in autumn) Table 6, indicates that interannual discharge variability in the Arno River basin is primarily driven by precipitation variability, consistent with the dominant role of precipitation seasonality and year-to-year rainfall fluctuations in Mediterranean flow dynamics (Merheb et al. 2016; Cid et al. 2017). The absence of significant streamflow trends is therefore coherent with the lack of precipitation trends over the study period. This contrasts with evidence from broader Mediterranean regions where long-term decreases in streamflow have been linked to warming-driven increases in evapotranspiration, compounded by declining precipitation (Masseroni et al. 2021).

Warm-season temperature–streamflow coupling

The negative summer T_{avg} –streamflow correlation indicates warm-season hydroclimatic sensitivity, but it does not demonstrate a single causal pathway. Higher summer temperature may coincide with lower streamflow through several interacting mechanisms, including reduced precipitation, enhanced atmospheric evaporative demand, antecedent storage deficits, irrigation and municipal abstractions, and reservoir operation. Because direct evapotranspiration, vapor-pressure deficit, soil moisture, groundwater, abstraction, and reservoir-operation data were not available in the present analysis, this relationship is interpreted as an observational association rather than proof of evaporative-demand control.

Implications for water resources management

The results of this study have implications for water-resource monitoring and planning in the Arno River basin, particularly because warming was most robust in summer and autumn while streamflow variability remained high. The large seasonal CV of streamflow, especially in autumn, indicates substantial year-to-year uncertainty in water availability that should be considered in reservoir operation, allocation planning, and drought preparedness. The small summer contribution to annual streamflow, together with the negative summer T_{avg} –streamflow correlation, identifies summer as a season requiring careful monitoring. However, whether continued warming will intensify low-flow conditions cannot be determined from the present observational correlation analysis alone and requires direct low-flow diagnostics, water-use information, and process-based modeling. The strong precipitation–streamflow coupling suggests that improved seasonal precipitation monitoring and forecasting could support operational water management.

Study limitations and future research directions

Several limitations should be noted, the 30-year record provides a recent climatological assessment but may not resolve longer multi-decadal variability, streamflow was analyzed at a single lower-basin hydrometric station, which cannot represent all sub-basin hydrological responses, water abstractions, reservoir operation, land-use changes, groundwater dynamics, soil moisture, vapor-pressure deficit, and direct evapotranspiration observations were not explicitly included; therefore, climate–streamflow associations should not be interpreted as direct causal mechanisms. IDW maps were generated using a distance-based method and should be interpreted as descriptive station-based precipitation surfaces rather than elevation-corrected precipitation

reconstructions, station homogeneity tests were used as diagnostic checks and do not replace full metadata-based relative homogenization. Future work should integrate longer homogenized records, multiple streamflow gauges, water-use and reservoir-operation data, direct land-surface diagnostics, low-flow and extreme-event indicators, and process-based hydrological modelling to better separate climatic, catchment-storage, and anthropogenic controls on streamflow.

Conclusions

This study analyzed hydroclimatic variability and trends in the Arno River basin, Tuscany, Italy, focusing on annual and seasonal patterns of temperature, precipitation, and streamflow from 1994 to 2023. The most robust signal was warming, especially in summer and autumn. Summer Tav_g increased by $+0.066\text{ }^{\circ}\text{C yr}^{-1}$ pBH=0.010, while autumn Tav_g increased by $+0.073\text{ }^{\circ}\text{C yr}^{-1}$ pBH=0.003, with both trends remaining significant after Benjamini–Hochberg correction. Annual Tav_g also increased by $+0.052\text{ }^{\circ}\text{C yr}^{-1}$ equivalent to about $+1.56\text{ }^{\circ}\text{C}$ over the study period, although this annual trend did not remain significant after multiple-testing correction. Change-point diagnostics identified a significant annual temperature shift around 2010, while station-level temperature diagnostics showed broadly coherent change points around the same period. These results support a basin-wide recent warming signal, particularly during the warm and transitional seasons. No statistically significant monotonic trends were detected for precipitation or streamflow at annual or seasonal scales. This should be interpreted as an absence of detectable monotonic trends within a short and highly variable 30-year record, not as evidence of hydrological stability. Interannual variability remained substantial, with CV values reaching 36.0% for summer precipitation and 65.8% for autumn streamflow. Spatially, precipitation showed a clear orographic gradient, with lower annual totals in the central and southwestern lowlands and higher totals along the Apennine ridge. Seasonal runoff was concentrated mainly in winter and spring, which together contributed approximately 76% of annual streamflow. Precipitation and streamflow were positively correlated across all seasons $\rho=0.51\text{--}0.93$, indicating strong precipitation–streamflow coupling in the basin. Summer Tav_g was negatively correlated with streamflow $\rho=-0.53$, $p=0.003$; pBH=0.009, indicating warm-season hydroclimatic sensitivity. However, this association should not be interpreted as proof of a single causal mechanism because summer streamflow may also be affected by rainfall deficits, antecedent storage, groundwater dynamics, abstractions, irrigation demand, and reservoir operation.

Overall, this observational synthesis provides an updated hydroclimatic baseline for the Arno River basin. The results highlight summer and autumn warming, strong seasonal precipitation–streamflow coupling, high streamflow variability, and the need for continued hydroclimatic monitoring. Future work should integrate longer homogenized records, multiple streamflow gauges, water-use and reservoir-operation data, direct land-surface diagnostics, low-flow indicators, and process-based hydrological modelling to better separate climatic, catchment-storage, and anthropogenic controls on streamflow.

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Author contributions Hafeez Ahmed Talpur: Writing—original draft, methodology, formal analysis, data curation, software, visualization, investigation, validation, writing—review and editing, conceptualization. Aziz Ahmed: Writing—review and editing, methodology, investigation, validation, software. Amanullah Mahar: Writing—review and editing, methodology, investigation, validation. Sadaf Nasreen: Writing—review and editing, methodology, investigation, data curation. Shakeel Ahmed Talpur: Writing—review and editing, methodology, investigation, software, visualization. Piero Di Carlo: Supervision, writing—review and editing, methodology, investigation, validation, funding acquisition, resources, project administration, conceptualization.

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Data availability Hydro-meteorological data can be accessed through the Regional Hydrological Service (SIR) of Tuscany portal.

Declarations

Conflict of interest The authors declare no competing interests.

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